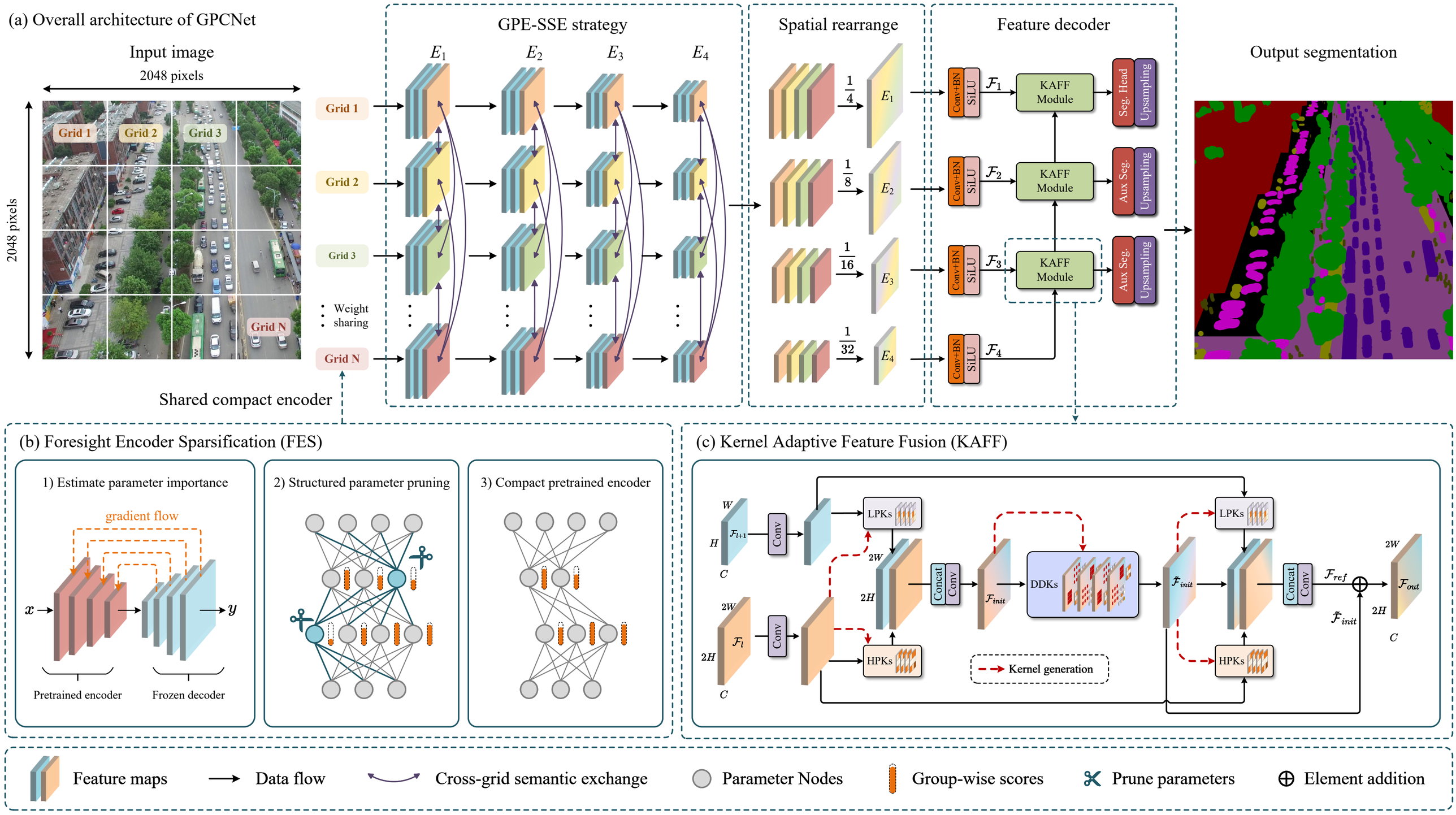

(a) Overall architecture of GPCNet
Input image
2048 pixels
2048 pixels
Grid 1
Grid 2
Grid 3
Grid N
GPE-SSE strategy
E_1
E_2
E_3
E_4
Weight sharing
Spatial rearrange
$\frac{1}{4}$
$\frac{1}{8}$
$\frac{1}{16}$
$\frac{1}{32}$
Feature decoder
Conv+BN
SiLU
$\mathcal{F}_1$
$\mathcal{F}_2$
$\mathcal{F}_3$
$\mathcal{F}_4$
KAFF Module
Seg. Head
Aux Seg.
Upsampling
Output segmentation
Shared compact encoder
(b) Foresight Encoder Sparsification (FES)
1) Estimate parameter importance
gradient flow
x
y
Pretrained encoder
Frozen decoder
2) Structured parameter pruning
3) Compact pretrained encoder
(c) Kernel Adaptive Feature Fusion (KAFF)
Conv
LPKs
HPKs
Concat
DDKs
$\mathcal{F}_{init}$
$\tilde{\mathcal{F}}_{init}$
$\mathcal{F}_{ref}$
$\mathcal{F}_{out}$
Kernel generation
Feature maps
Data flow
Cross-grid semantic exchange
Parameter Nodes
Group-wise scores
Prune parameters
Element addition